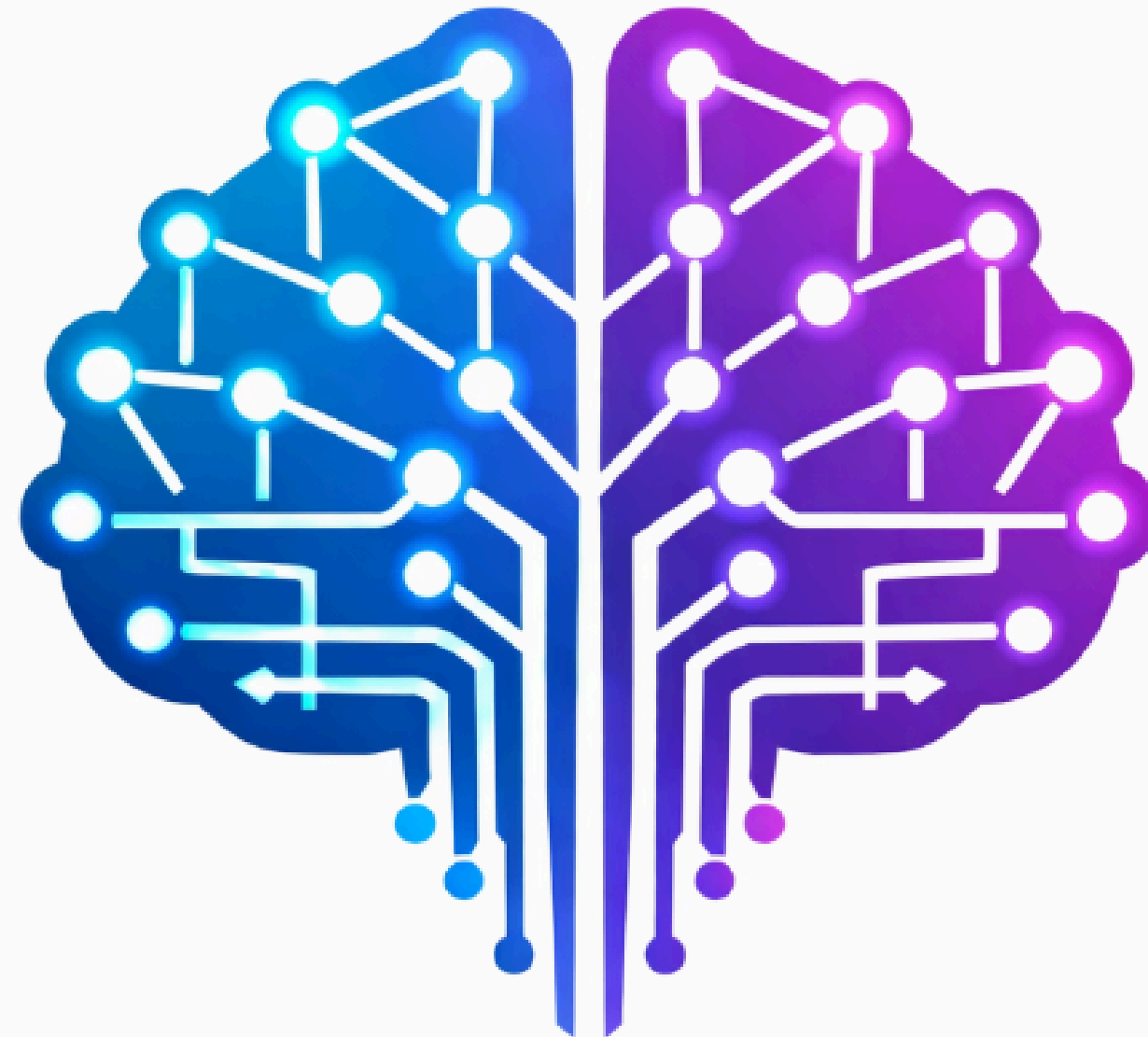


GENAI FUNDAMENTALS



LECTURE:
LLM ECOSYSTEM & MODEL LANDSCAPE

The Rapidly Growing LLM Ecosystem

The Generative AI ecosystem is expanding rapidly.

Today's ecosystem includes:

- Model providers
- Cloud platforms
- Open-source communities
- Developer tools

👉 LLMs are becoming a **core layer of modern software systems.**

Major LLM Providers

Some major organizations building LLMs include:

Closed Model Providers

- OpenAI
- Anthropic
- Google

Open Model Contributors

- Meta
- Mistral
- Stability AI

👉 These organizations drive the evolution of modern LLMs.

Closed Models vs Open Models

Closed Models

- Access through APIs
- Training data and weights not public
- Managed by providers

Examples: GPT, Claude, Gemini

Open Models

- Model weights available
- Can run locally or on private servers

Examples: LLaMA, Mistral, Falcon

API-Based Models

Most developers interact with LLMs through **APIs**.

Typical workflow:

Application → API request → LLM → Generated response

Benefits:

- easy integration
- scalable infrastructure
- no model hosting required

Local & Self-Hosted Models

Developers can also run **LLMs locally or on private infrastructure**.

Reasons to run local models:

- data privacy
- offline usage
- cost control
- customization

Common tools:

- Ollama
- Hugging Face
- vLLM
- LM Studio

The LLM Stack for Developers

☞ A typical GenAI application stack includes:

User Interface



Application Backend



LLM / Model API



Retrieval Systems / Databases



External Tools & APIs

LLMs are one component of a larger system architecture.

How Developers Choose an LLM

👉 When selecting an LLM, developers consider:

- model capability
- latency
- cost
- context window
- privacy requirements
- deployment options

👉 Choosing the right model is an **engineering decision**.

